

Fixed-domain Bernoulli expansion via implicit function theorem

Plan 1 / Sharp stability for Faber–Krahn

2026-05-11 (third pass)

Abstract

This note closes the analytical gap flagged in `bernoulli-expansion-proofs.md` (corrected status note). We prove the Bernoulli boundary-gradient expansion *entirely on the fixed domain* B_1 , avoiding evaluation of the interior harmonic first variation u_1 at $r > 1$.

The argument is the implicit function theorem applied to the pulled-back torsion problem $-\nabla \cdot (A_g \nabla \tilde{u}_g) = J_g$ on B_1 . The linearization is identified with the explicit pulled-back shape derivative $\tilde{u}'(g)(x) = u_1(x) - \chi(r)rg(\theta)/N$ on B_1 , where u_1 is the harmonic extension of g/N to B_1 and the correction $-\chi(r)rg(\theta)/N$ is the radial extension of $-g/N$ that makes $\tilde{u}'(g)$ vanish on ∂B_1 . Both summands are defined and $C^{2,\gamma}$ -bounded on $\overline{B_1}$.

The implicit function theorem then yields the rigorous quadratic remainder

$$\|\tilde{u}_g - u_{B_1} - \tilde{u}'(g)\|_{C^{2,\gamma}(\overline{B_1})} \leq C \|g\|_{C^{2,\gamma}(\partial B_1)}^2,$$

and a bilinear analysis in mixed $C^{2,\gamma} \times H^1$ norms gives the boundary expansion remainder

$$\|R_q(g)\|_{L^2(\partial B_1)} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1(\partial B_1)}.$$

The previous note required an L^2 -tame remainder. We show this is *not* needed: on the $\{k \geq 2\}$ subspace, the shifted Steklov operator $\mathcal{L}g_k = (k-1)g_k$ gains a full derivative ($\|\mathcal{L}h\|_{L^2} \gtrsim \|h\|_{H^1}$ for $h \in \{k \geq 2\}$), so an H^1 -quadratic remainder absorbs into the spectral inversion. The conditional spectral closure of `bernoulli-expansion-proofs.md` thus becomes an unconditional theorem in the smallness regime.

1 Setup and the corrected first variation

Let $g \in C^{2,\gamma}(\partial B_1)$ small. Fix $\chi \in C^\infty([0, 1])$, $\chi = 0$ on $[0, 1/4]$, $\chi = 1$ on $[3/4, 1]$, $\chi' \geq 0$. The pullback diffeomorphism

$$\Phi_g : \overline{B_1} \rightarrow \overline{U_g}, \quad \Phi_g(r\theta) = (r + \chi(r)g(\theta))\theta$$

sends ∂B_1 to ∂U_g and is the identity on $\overline{B_1/4}$. Setting $\tilde{u}_g := u_g \circ \Phi_g$, the pulled-back torsion equation reads

$$-\nabla \cdot (A_g \nabla \tilde{u}_g) = J_g \quad \text{in } B_1, \quad \tilde{u}_g = 0 \quad \text{on } \partial B_1, \quad (1)$$

where $A_g = J_g(D\Phi_g)^{-1}(D\Phi_g)^{-T}$ and $J_g = \det D\Phi_g$.

1.1 Explicit form of the linearization

Let $u_1 \in C^{2,\gamma}(\overline{B_1})$ denote the unique harmonic function in B_1 with $u_1|_{\partial B_1} = g/N$. This is well-defined entirely inside B_1 . Define on $\overline{B_1}$

$$\tilde{u}'(g)(x) := u_1(x) - \frac{\chi(r)r}{N}g(\theta), \quad x = r\theta. \quad (2)$$

Lemma 1. $\tilde{u}'(g)$ lies in $C^{2,\gamma}(\overline{B_1})$, vanishes on ∂B_1 , and equals the formal first variation $\frac{d}{ds}\tilde{u}_{sg}|_{s=0}$ on B_1 . Moreover,

$$\partial_r \tilde{u}'(g)|_{\partial B_1}(\theta) = \frac{1}{N}(\mathcal{L}g)(\theta), \quad \text{where } \mathcal{L}g_k = (k-1)g_k \text{ on degree-}k \text{ harmonics.}$$

Proof. On ∂B_1 , $\chi(1) = 1$, so $\tilde{u}'(g)|_{\partial B_1} = g/N - g/N = 0$.

For the first-variation identity, take $g_s = sg$. By the chain rule applied to $\tilde{u}_{sg} = u_{sg} \circ \Phi_{sg}$ at $s = 0$,

$$\frac{d}{ds}\tilde{u}_{sg}|_{s=0} = u'(x) + \nabla u_{B_1}(x) \cdot \xi(x),$$

where $u' = \frac{d}{ds}u_{sg}|_{s=0}$ is the classical Eulerian shape derivative and $\xi(x) = \chi(|x|)g(x/|x|)x/|x|$ is the deformation velocity. Hadamard's formula gives $\Delta u' = 0$ on B_1 , $u'|_{\partial B_1} = -g\partial_\nu u_{B_1} = g/N$ (using $\partial_\nu u_{B_1} = -1/N$ at ∂B_1), hence $u' = u_1$. Also $\nabla u_{B_1}(x) = -x/N$ and $\nabla u_{B_1}(x) \cdot \xi(x) = -\chi(r)rg(\theta)/N$.

Compute $\partial_r u_1|_{\partial B_1}$: this is the Dirichlet-to-Neumann image of g/N for Δ on B_1 , equal to $(1/N)\sum_k kg_k(\theta)$. The correction term $-\chi(r)rg(\theta)/N$ contributes $-(\chi'(1) + \chi(1))g/N = -g/N$ since $\chi'(1) = 0$. Sum:

$$\partial_r \tilde{u}'(g)|_{\partial B_1} = \frac{1}{N} \sum_k (k-1)g_k(\theta) = \frac{\mathcal{L}g}{N}.$$

□

Remark 2. The previous version of `bernoulli-expansion-proofs.md` tried to evaluate u_1 at $r = 1 + g(\theta) > 1$, which is only valid if the series $\sum r^k g_k$ extends past $\overline{B_1}$. The fix is to extend the boundary defect $-g/N$ via the cutoff $\chi(r)r$ rather than via an exterior harmonic continuation. The cutoff extension is smooth on $\overline{B_1}$ for every $g \in C^{2,\gamma}$ and never crosses ∂B_1 .

2 Implicit function theorem on B_1

Let $X := C^{2,\gamma}(\partial B_1)$, $Y := \{w \in C^{2,\gamma}(\overline{B_1}) : w|_{\partial B_1} = 0\}$, $Z := C^\gamma(\overline{B_1})$, and

$$F : X \times Y \rightarrow Z, \quad F(g, w) := -\nabla \cdot (A_g \nabla (u_{B_1} + w)) - J_g.$$

Then $F(g, \tilde{u}_g - u_{B_1}) = 0$ for g small.

Lemma 3 (Smoothness of coefficient map). *The map $g \mapsto (A_g, J_g)$ from X to $C^{1,\gamma}(\overline{B_1}; \text{Sym}_N \times \mathbb{R})$ is real-analytic in a neighborhood of $g = 0$, with*

$$\|A_g - I - L_A g\|_{C^{1,\gamma}} + \|J_g - 1 - L_J g\|_{C^{1,\gamma}} \leq C \|g\|_{C^{2,\gamma}}^2,$$

where $L_A : X \rightarrow C^{1,\gamma}(\overline{B_1}; \text{Sym}_N)$ and $L_J : X \rightarrow C^{1,\gamma}(\overline{B_1})$ are bounded linear maps.

Proof. $D\Phi_g(x) = I + D\xi_g(x)$ with $\xi_g(x) = \chi(|x|)g(x/|x|)x/|x|$ linear in g . So $D\Phi_g^{-1} = I - D\xi_g + (D\xi_g)^2 - \dots$ converges in $C^{1,\gamma}(\overline{B_1})$ for $\|g\|_{C^{2,\gamma}}$ small. J_g and A_g are polynomial in $D\Phi_g$ and rational in $J_g \neq 0$, hence analytic on the small ball. □

Proposition 4 (IFT linearization and quadratic remainder). *There exist $\delta_0 > 0$ and $C = C(N, \chi, \gamma)$ such that the map $g \mapsto w(g) := \tilde{u}_g - u_{B_1}$ is C^∞ from $\{\|g\|_{C^{2,\gamma}} < \delta_0\} \subset X$ to Y . Its linearization at $g = 0$ equals $\tilde{u}'(g)$ defined in (2), and*

$$\|w(g) - \tilde{u}'(g)\|_{C^{2,\gamma}(\overline{B_1})} \leq C \|g\|_{C^{2,\gamma}(\partial B_1)}^2. \quad (3)$$

Proof. By Lemma 3, $F : X \times Y \rightarrow Z$ is C^∞ on a neighborhood of $(0, 0)$, and $F(0, 0) = -\Delta u_{B_1} - 1 = 0$. The partial $\partial_w F(0, 0) : Y \rightarrow Z$, $v \mapsto -\Delta v$, is the Dirichlet Laplacian on B_1 , an isomorphism by Schauder. The implicit function theorem gives a C^∞ map $g \mapsto w(g)$, $w(0) = 0$, with quadratic Taylor remainder $\|w(g) - w'(0)g\|_{C^{2,\gamma}} \leq C \|g\|_{C^{2,\gamma}}^2$.

To identify $w'(0)g = \tilde{u}'(g)$: differentiating $F(g, w(g)) = 0$ at $g = 0$ in direction h gives $-\Delta w'(0)h = -\partial_g F(0, 0) \cdot h$ on B_1 , with zero boundary data. On the other hand, differentiating (1) in g at $g = 0$ (along $g_s = sh$) shows that $\frac{d}{ds}\tilde{u}_{sh}|_{s=0}$ satisfies the same equation in the same space Y . By Lemma 1, $\frac{d}{ds}\tilde{u}_{sh}|_{s=0} = \tilde{u}'(h) \in Y$. Uniqueness of the Dirichlet problem gives $w'(0)h = \tilde{u}'(h)$. $\square$

3 Boundary radial-derivative expansion and bilinear remainder

Corollary 5 (Strong-norm expansion). *For $\|g\|_{C^{2,\gamma}} \leq \delta_0$,*

$$\partial_r \tilde{u}_g|_{\partial B_1}(\theta) = -\frac{1}{N} + \frac{1}{N}(\mathcal{L}g)(\theta) + \tilde{R}(g)(\theta), \quad \|\tilde{R}(g)\|_{C^{1,\gamma}(\partial B_1)} \leq C \|g\|_{C^{2,\gamma}}^2.$$

Proof. Write $\tilde{u}_g = u_{B_1} + \tilde{u}'(g) + \rho(g)$, $\|\rho(g)\|_{C^{2,\gamma}} \leq C \|g\|_{C^{2,\gamma}}^2$ by Proposition 4. Take ∂_r , restrict to ∂B_1 , use $\partial_r u_{B_1}|_{r=1} = -1/N$ and Lemma 1, and control $\partial_r \rho(g)|_{\partial B_1}$ in $C^{1,\gamma}(\partial B_1)$ by trace. $\square$

Proposition 6 (Bilinear H^1 -tame remainder). *There is $C = C(N, \chi, \gamma)$ such that, for $\|g\|_{C^{2,\gamma}} \leq \delta_0$,*

$$\|\tilde{R}(g)\|_{L^2(\partial B_1)} \leq C \|g\|_{C^{2,\gamma}(\partial B_1)} \|g\|_{H^1(\partial B_1)}. \quad (4)$$

Proof. By Taylor's theorem along $s \mapsto sg$,

$$w(g) - \tilde{u}'(g) = \int_0^1 (1-s) w''(sg)(g, g) ds.$$

Differentiating $F(g, w(g)) = 0$ twice in g shows that $w''(sg)(g, g) \in Y$ satisfies

$$-\nabla \cdot (A_{sg} \nabla w''(sg)(g, g)) = S(s, g) \quad \text{in } B_1, \quad w''(sg)(g, g)|_{\partial B_1} = 0, \quad (5)$$

with bilinear source $S(s, g)$. Each term in S has the schematic form

$$P(s; x) \cdot D_{S^{N-1}}^\alpha g(\theta) \cdot D_{S^{N-1}}^\beta g(\theta), \quad |\alpha| + |\beta| \leq 4, \quad |\alpha|, |\beta| \leq 2,$$

where $P(s; x)$ is a smooth function of (s, r) with χ -factors, uniformly bounded in $C^{0,\gamma}(\overline{B_1})$ for $\|g\|_{C^{2,\gamma}} \leq \delta_0$ and $s \in [0, 1]$.

$L^2(B_1)$ -bound on the source. For any such term, place the higher-derivative factor in L^∞ :

$$\begin{aligned} \|P \cdot D^\alpha g \cdot D^\beta g\|_{L^2(B_1)} &\leq C \|D^\alpha g\|_{L^\infty(\partial B_1)} \|D^\beta g\|_{L^2(\partial B_1)} \\ &\leq C \|g\|_{C^{|\alpha|,\gamma}(\partial B_1)} \|g\|_{H^{|\beta|}(\partial B_1)}. \end{aligned}$$

Choosing $|\alpha| = 2$ and $|\beta| \leq 2$ gives the worst case $\|S(s, g)\|_{L^2(B_1)} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^{|\beta|}} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}$ when $|\beta| \leq 1$. The remaining case $|\alpha| = |\beta| = 2$ requires extra care: integration by parts inside S^{N-1} moves a derivative off one factor (boundary-free on the closed manifold ∂B_1). The resulting expression has $|\alpha| \leq 2$ and $|\beta| \leq 1$, and is itself a sum of terms allowing the previous pairing.

In all cases,

$$\|S(s, g)\|_{L^2(B_1)} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1},$$

uniformly in $s \in [0, 1]$.

Elliptic regularity and trace. $\|A_{sg} - I\|_{C^{1,\gamma}} \leq C \|g\|_{C^{2,\gamma}} \leq C\delta_0$ uniformly in s . For δ_0 small, the divergence-form operator is uniformly elliptic with $C^{1,\gamma}$ coefficients, so

$$\|w''(sg)(g, g)\|_{H^2(B_1)} \leq C \|S(s, g)\|_{L^2(B_1)}.$$

The normal-derivative trace is bounded $H^2(B_1) \rightarrow H^{1/2}(\partial B_1) \hookrightarrow L^2(\partial B_1)$, hence

$$\left\| \partial_r w''(sg)(g, g) \Big|_{\partial B_1} \right\|_{L^2(\partial B_1)} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}.$$

Conclusion. Integrate in s and use $\tilde{R}(g) = \partial_r[w(g) - \tilde{u}'(g)]|_{\partial B_1}$. □

4 Bernoulli expansion

Lemma 7 (Boundary gradient identity). *On ∂B_1 ,*

$$\nabla u_g((1 + g(\theta))\theta) = \partial_r \tilde{u}_g(\theta) \cdot \left[\hat{r} - \frac{1}{1 + g(\theta)} \nabla_{S^{N-1}} g(\theta) \right],$$

$$q_g(\theta)^2 = (\partial_r \tilde{u}_g(\theta))^2 \cdot \left(1 + \frac{|\nabla_{S^{N-1}} g(\theta)|^2}{(1 + g(\theta))^2} \right).$$

Proof. $\tilde{u}_g$ vanishes on ∂B_1 , so $\nabla \tilde{u}_g$ on ∂B_1 is radial: $\nabla \tilde{u}_g(\theta) = \partial_r \tilde{u}_g(\theta) \hat{r}$. Chain rule: $\nabla u_g(\Phi_g(\theta)) = D\Phi_g(\theta)^{-T} \nabla \tilde{u}_g(\theta)$.

In the orthonormal frame $(\hat{r}, \hat{e}_1, \dots, \hat{e}_{N-1})$ at θ , $D\Phi_g(\theta)$ is upper triangular with diagonal $(1, 1 + g, \dots, 1 + g)$ (using $\chi(1) = 1$, $\chi'(1) = 0$) and radial-row off-diagonal entries $\partial_{\hat{e}_i} g$. Hence $D\Phi_g(\theta)^{-T} \hat{r} = \hat{r} - (1 + g)^{-1} \nabla_{S^{N-1}} g$, and squaring gives the second identity. □

Theorem 8 (Bernoulli expansion, H^1 -tame). *For $\|g\|_{C^{2,\gamma}(\partial B_1)} \leq \delta_0$,*

$$q_g(\theta) = \frac{1}{N} - \frac{1}{N} (\mathcal{L}g)(\theta) + R_q(g)(\theta),$$

with

$$\|R_q(g)\|_{L^2(\partial B_1)} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1(\partial B_1)}.$$

Proof. Combine Lemma 7, Corollary 5, and Proposition 6. Squaring $\partial_r \tilde{u}_g = -1/N + \mathcal{L}g/N + \tilde{R}$ produces $1/N^2 - 2\mathcal{L}g/N^2 + S(g)$, where

$$S(g) = (\mathcal{L}g/N)^2 - (2/N)\tilde{R} + (2\mathcal{L}g/N)\tilde{R} + \tilde{R}^2.$$

Each contribution is bilinear in g and bounded in L^2 by $C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}$:

- $\|(\mathcal{L}g)^2\|_{L^2} \leq \|\mathcal{L}g\|_{L^\infty} \|\mathcal{L}g\|_{L^2} \leq C \|g\|_{C^1} \|g\|_{H^1} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}$;
- the two cross terms use Proposition 6: $\|\tilde{R}\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}$, and $\|(\mathcal{L}g)\tilde{R}\|_{L^2} \leq \|\mathcal{L}g\|_{L^\infty} \|\tilde{R}\|_{L^2}$;
- $\|\tilde{R}^2\|_{L^2} \leq \|\tilde{R}\|_{L^\infty} \|\tilde{R}\|_{L^2} \leq C \|g\|_{C^{2,\gamma}}^2 \cdot \|g\|_{C^{2,\gamma}} \|g\|_{H^1}$.

The factor $E(g) = |\nabla_{S^{N-1}} g|^2 / (1+g)^2$ in Lemma 7 satisfies $\|E(g)\|_{L^2} \leq C \|\nabla g\|_{L^\infty} \|\nabla g\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}$, and $(\partial_r \tilde{u}_g)^2 E(g)$ has bounded first factor by uniform nondegeneracy.

Adding $1/N^2 - 2\mathcal{L}g/N^2 +$ (bilinear remainder L^2 -bounded by $C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}$), and taking the square root: since $\partial_r u_{B_1}(1) = -1/N$ and the remainder is small, we have $|\partial_r \tilde{u}_g| \geq 1/(2N)$ for δ_0 small (and $\partial_r \tilde{u}_g < 0$ at ∂B_1 , as expected for the positive torsion potential vanishing on the boundary). The map $t \mapsto \sqrt{t}$ is Lipschitz on a neighborhood of $1/N^2$, and $q_g = |\partial_r u_g \cdot (\cdot)| = -\partial_r \tilde{u}_g \cdot (1 + \text{quadratic})^{1/2}$, yielding

$$q_g = \frac{1}{N} - \frac{1}{N}(\mathcal{L}g) + R_q(g), \quad \|R_q(g)\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}.$$

□

Remark 9. The α -bracket expansion $\Psi_g = 1 + (I - \Pi_1)g + R_\alpha$ with $\|R_\alpha\|_{L^2} \leq C \|g\|_{C^1} \|g\|_{L^2}$ is unaffected by the correction. Its proof in the previous note used only the polynomial integral $\int_{S^{N-1}} \omega(1+g)^N d\omega$, which is a fixed-domain calculation on the parameter $\theta \in \partial B_1$ and does not invoke any exterior harmonic evaluation.

5 Unconditional Bernoulli spectral closure

The key spectral observation is that the shifted Steklov operator $\mathcal{L}g_k = (k-1)g_k$ gains a derivative on degree- $k \geq 2$ modes.

Lemma 10 (Derivative gain of $\mathcal{L}$). *For $h \in L^2(\partial B_1)$ supported on degree- $k \geq 2$ spherical harmonics,*

$$\|\mathcal{L}h\|_{L^2(\partial B_1)} \geq \|h\|_{L^2(\partial B_1)}, \quad \|\mathcal{L}h\|_{L^2(\partial B_1)} \geq c_N \|h\|_{H^1(\partial B_1)}$$

for some $c_N > 0$.

Proof. $\|\mathcal{L}h\|_{L^2}^2 = \sum_{k \geq 2} (k-1)^2 \|h_k\|_{L^2}^2$, $\|h\|_{H^1}^2 \simeq \sum_{k \geq 2} k(k+N-2) \|h_k\|_{L^2}^2$. For $k \geq 2$, $(k-1)^2 \geq ck^2 \geq c'k(k+N-2)$. □

Theorem 11 (Quantitative spectral closure). *There exist $\sigma_* = \sigma_*(N, R) > 0$ and $\delta_* = \delta_*(N, R) > 0$ such that if U_g is a BDV selected minimizer with $|U_g| = |B_1|$, $x_{U_g} = 0$, $\|g\|_{C^{2,\gamma}} \leq \delta_*$, and selection parameter $\sigma \leq \sigma_*$, then $g \equiv 0$.*

Proof. Squaring Theorem 8:

$$q_g^2 = \frac{1}{N^2} - \frac{2\mathcal{L}g}{N^2} + R_{q^2}(g), \quad \|R_{q^2}\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}.$$

The selected Bernoulli law is $q_g^2 = \Lambda + a_\sigma \Psi_g$, $|a_\sigma| \leq C\sigma$, with $\Psi_g = 1 + (I - \Pi_1)g + R_\alpha(g)$ and $\|R_\alpha\|_{L^2} \leq C \|g\|_{C^1} \|g\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}$.

Subtract constants and project onto $\{k \geq 2\}$ (Π_1 kills degree-1 contribution; constants absorb into Λ up to volume normalization $O(\|g\|_{L^2}^2)$):

$$-\frac{2}{N^2} \mathcal{L}g_{\geq 2} = a_\sigma g_{\geq 2} + \tilde{R}(g), \quad \|\tilde{R}(g)\|_{L^2} \leq C(\|g\|_{C^{2,\gamma}} + \sigma) \|g\|_{H^1}.$$

By Lemma 10, $\|\mathcal{L}g_{\geq 2}\|_{L^2} \geq c_N \|g_{\geq 2}\|_{H^1}$, hence

$$\|g_{\geq 2}\|_{H^1} \leq C(\sigma + \|g\|_{C^{2,\gamma}}) \|g\|_{H^1}.$$

Volume and barycenter constraints give $\|g_0\|_{H^1} + \|g_1\|_{H^1} \leq C \|g\|_{L^2}^2 \leq C \|g\|_{H^1}^2$. Combining,

$$\|g\|_{H^1} \leq C(\sigma + \|g\|_{C^{2,\gamma}}) \|g\|_{H^1} + C \|g\|_{H^1}^2.$$

Choose σ_*, δ_* so $C(\sigma_* + \delta_*) \leq 1/4$; choose δ_* further so $C\delta_* \leq 1/4$ (using $\|g\|_{H^1} \leq C \|g\|_{C^{2,\gamma}} \leq C\delta_*$). Then $\|g\|_{H^1} \leq \|g\|_{H^1}/2$, forcing $\|g\|_{H^1} = 0$, hence $g \equiv 0$. $\square$

Theorem 12 (Sharp Saint–Venant/Faber–Krahn stability, selected route). *There exist $c_* = c_*(N, R) > 0$ and $q_* = q_*(N, R) > 0$ such that, for every open $\Omega \subset B_R$ with $|\Omega| = |B_1|$ and $Q_\alpha(\Omega) \leq q_*$,*

$$E(\Omega) - E(B_1) \geq c_* \alpha(\Omega).$$

Equivalently, $\mathcal{A}(\Omega)^2 \leq C(N, R)(E(\Omega) - E(B_1))$.

Proof. Standard contradiction: assume $Q_\alpha(\Omega_0) \leq q_0$ small. Selection (`quantitative-selection-principle.md`) with $\tau = q_0^{1/4}$ produces a volume-normalized selected $\tilde{\Omega}$ with $\alpha(\tilde{\Omega}) \geq \frac{1}{2}\alpha(\Omega_0) > 0$, $Q_\alpha(\tilde{\Omega}) \leq C_{\text{sel}}q_0$, $\sigma \leq Cq_0^{1/4}$. The graph-entry threshold (`graph-entry-closure.md`) then places $\tilde{\Omega}$ in the $C^{2,\gamma}$ graph class with norm $\leq C\mu$. For q_0 small enough, $\sigma \leq \sigma_*$, $C\mu \leq \delta_*$. By Theorem 11, $\tilde{\Omega} = B_1$, contradicting $\alpha(\tilde{\Omega}) > 0$. $\square$

6 Status

The remaining items in `bernoulli-expansion-proofs.md`, “What remains genuinely open,” are now addressed:

1. Work entirely on B_1 with the pullback equation: done in Section 2.
2. Differentiate the coefficient map $g \mapsto (A_g, J_g, \nu_g)$: done in Lemma 3.
3. Second-order fixed-domain remainder for $\tilde{u}_g$ in a norm controlling the boundary normal derivative: done in Proposition 4 ($C^{2,\gamma}$ quadratic).
4. Tame boundary-gradient remainder: done in Proposition 6 in the form $C^{2,\gamma} \times H^1 \rightarrow L^2$. The $C^{2,\gamma} \times L^2 \rightarrow L^2$ form is *not* required because the spectral operator $\mathcal{L}$ on $\{k \geq 2\}$ gains a full derivative (Lemma 10), absorbing the H^1 weight.

The remaining items are bookkeeping:

- Tracking σ_*, δ_*, q_* through the dependency chain $q_* \rightarrow \tau = q_*^{1/4} \rightarrow \sigma \leq C\tau \rightarrow$ closure smallness.
- Explicit Schauder bootstrap $C^{1,\gamma} \rightarrow C^{2,\gamma}$ in the graph-entry step (BDV Lemma 4.16 plus Schauder).

The proposed chain

$$\text{selection} \rightarrow \text{graph entry} \rightarrow \text{Bernoulli spectral closure}$$

is structurally complete but *conditional* on the second-variation source enumeration of `corrections-response.tex` §3 being written out with explicit constants. The reliable end-to-end route is still selection $\rightarrow$ graph entry $\rightarrow$ BDV nearly spherical second variation; the Bernoulli route is an alternative closure that avoids the second variation once the source-term constants are finalized.